Name:_____

Ionic Formula

A formula tells you how many atoms of each element are in one molecule of a compound.

Formula	Number and type of element
NaCl	1 sodium 1 chlorine
H_2O	___ hydrogen ___ oxygen
$PbCl_5$	
Fe_2O_3	
Na_2O	
CaO	
$FeCl_3$	
Fe_2O_3	

What ion is always written first?_____

If you are given the name of an ionic molecule you can work out its formula

If both ions have the same charge the charges cancel out.

Name	Ions	formula
Postassium iodide	K^+ I^-	KI
Calcium oxide	Ca^{+2} O^{-2}	CaO

What do you notice about the sound at the end of the non-metal ion?

If the ions have different charges we remove the + or – sign and “cross” the numbers over.

Calcium Iodide Ca^{+2} I^{-}

CaI_2

Sodium Oxide Na^{+} O^{-2}

Aluminium Oxide

Choose one of your own:

Cations		Anions	
Name	Formula	Name	Formula
Hydrogen	H^+	Hydroxide	OH^-
Sodium	Na^+	Chloride	Cl^-
Potassium	K^+	Nitrate	NO_3^-
Ammonium	NH_4^+	Acetate	CH_3COO^-
Silver	Ag^+	Bicarbonate	HCO_3^-
Calcium	Ca^{2+}	Sulfide	S^{2-}
Iron(II)	Fe^{2+}	Oxide	O^{2-}
Copper	Cu^{2+}	Sulfate	SO_4^{2-}
Lead	Pb^{2+}	Carbonate	CO_3^{2-}
Iron(III)	Fe^{3+}		
Aluminum	Al^{3+}		